Crank–Nicolson, the θ-scheme, and Rannacher time-stepping

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1 Setting and motivation

FTCS and BTCS (previous two notes) are first-order in time: their local truncation error is $O(\Delta\tau)+O(\Delta x^2)$. To balance the spatial and temporal contributions one needs $\Delta\tau \sim \Delta x^2$, which forces $M \sim N^2$ and total cost $O(N^3)$ for both schemes (FTCS by CFL, BTCS by accuracy).

This note implements *Crank–Nicolson* (CN), the unique scheme in the convex family of FTCS and BTCS that is *second-order in time*: $O(\Delta\tau^2) + O(\Delta x^2)$. With $M \sim N$, CN achieves $O(\Delta x^2)$ precision at $O(N^2)$ total cost — a factor-N improvement over FTCS/BTCS in the same precision regime. CN is the workhorse of finite-difference option pricers in production.

The deliverable is a θ-scheme infrastructure (`quantlib.theta_scheme`) that subsumes FTCS, BTCS and CN as parameter choices, plus the CN pricer with Rannacher time-stepping (`quantlib.cn`) to suppress spurious oscillations near the strike kink.

2 The θ-family

The three schemes FTCS, BTCS and CN are particular cases of a single parametric family:

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} \;=\; \theta\,\mathcal{L}_h V_j^{n+1} \;+\; (1-\theta)\,\mathcal{L}_h V_j^n, \qquad \theta \in [0,1], \tag{1}$$

where $\mathcal{L}_h$ is the discrete spatial operator

$$\mathcal{L}_h V_j \;=\; \frac{\sigma^2}{2}\,\frac{V_{j+1} - 2V_j + V_{j-1}}{\Delta x^2} + \mu\,\frac{V_{j+1} - V_{j-1}}{2\Delta x} - r\,V_j.$$

The cases of interest are

θ	Name	Type	Stability	Time order
0	FTCS	Explicit	Conditional ($\alpha \le 1/2$)	$O(\Delta\tau)$
1/2	**Crank–Nicolson**	Implicit	Unconditional	$O(\Delta\tau^2)$
1	BTCS	Implicit	Unconditional	$O(\Delta\tau)$

The order jump at $\theta = 1/2$ is the central observation of this note. It is the pay-off of the symmetry between the new and old levels: when the right-hand side is the average of $\mathcal{L}_h V$ at τ_n and τ_{n+1}, the discrete equation is centred at the *midpoint* $\tau_{n+1/2}$, and the midpoint rule is one order more accurate than the endpoint rules.

3 Crank–Nicolson explicitly

Setting $\theta = 1/2$ in (1), expanding $\mathcal{L}_h$, and rearranging V^{n+1} to the left, V^n to the right:

$$\boxed{\beta_-\,V_{j-1}^{n+1} + \beta_0\,V_j^{n+1} + \beta_+\,V_{j+1}^{n+1} \;=\; \gamma_-\,V_{j-1}^n + \gamma_0\,V_j^n + \gamma_+\,V_{j+1}^n.} \tag{2}$$

With $\alpha = \frac{\sigma^2}{2}\,\Delta\tau/\Delta x^2$ and $\nu = \mu\,\Delta\tau/(2\Delta x)$ as before, the LHS coefficients are

$$\beta_- = -\tfrac{1}{2}\alpha + \tfrac{1}{2}\nu, \quad \beta_0 = 1 + \alpha + \tfrac{1}{2}r\Delta\tau, \quad \beta_+ = -\tfrac{1}{2}\alpha - \tfrac{1}{2}\nu,$$

and the RHS coefficients are

$$\gamma_- = +\tfrac{1}{2}\alpha - \tfrac{1}{2}\nu, \quad \gamma_0 = 1 - \alpha - \tfrac{1}{2}r\Delta\tau, \quad \gamma_+ = +\tfrac{1}{2}\alpha + \tfrac{1}{2}\nu.$$

Symmetry: $\beta_\pm = -\frac{1}{2}$(the corresponding BTCS coefficient minus 1 on diagonal), $\gamma_\pm = +\frac{1}{2}$(the corresponding F
CN is, structurally, "half BTCS plus half FTCS".

4 Why CN is second-order in time: the midpoint rule

The non-trivial claim is that the local truncation error of (2) is $O(\Delta\tau^2) + O(\Delta x^2)$, despite the fact that the time difference $(V^{n+1} - V^n)/\Delta\tau$ looks like the same first-order forward Euler quotient that FTCS uses. The gain comes from *which time level the equation is centred on.*

Expand V around the midpoint $\tau_{n+1/2} = \tau_n + \Delta\tau/2$:

$$V(\tau_{n+1}) = V_* + \tfrac{1}{2}\Delta\tau\,V_*' + \tfrac{1}{8}\Delta\tau^2\,V_*'' + \tfrac{1}{48}\Delta\tau^3\,V_*''' + O(\Delta\tau^4),$$
$$V(\tau_n) = V_* - \tfrac{1}{2}\Delta\tau\,V_*' + \tfrac{1}{8}\Delta\tau^2\,V_*'' - \tfrac{1}{48}\Delta\tau^3\,V_*''' + O(\Delta\tau^4),$$

where $V_* := V(\tau_{n+1/2})$, $V_*' := V_\tau(\tau_{n+1/2})$, etc. Subtracting and dividing by $\Delta\tau$:

$$\frac{V(\tau_{n+1}) - V(\tau_n)}{\Delta\tau} \;=\; V_\tau(\tau_{n+1/2}) \;+\; \frac{\Delta\tau^2}{24}\,V_{\tau\tau\tau}(\tau_{n+1/2}) \;+\; O(\Delta\tau^4). \tag{3}$$

The $\Delta\tau$ term *cancels by symmetry*: the quotient is a second-order approximation to V_τ at the midpoint.

Analogously, adding the two expansions of $\mathcal{L}_h V$ (one at τ_n, one at τ_{n+1}) and dividing by 2:

$$\tfrac{1}{2}[\mathcal{L}_h V(\tau_{n+1}) + \mathcal{L}_h V(\tau_n)] \;=\; \mathcal{L}_h V(\tau_{n+1/2}) \;+\; O(\Delta\tau^2) \;+\; O(\Delta x^2), \tag{4}$$

where the $O(\Delta\tau^2)$ comes from the time interpolation (linear average of the endpoints) and the $O(\Delta x^2)$ is the spatial truncation of $\mathcal{L}_h$. Combining (3) and (4), the CN equation $(V^{n+1} - V^n)/\Delta\tau = \frac{1}{2}(\mathcal{L}_h V^{n+1} + \mathcal{L}_h V^n)$ is the discretisation of the exact PDE $V_\tau = \mathcal{L}V$ *at the midpoint*, with truncation

$$\tau_h = O(\Delta\tau^2) + O(\Delta x^2).$$

FTCS and BTCS, by contrast, evaluate the right-hand side at τ_n or τ_{n+1} (not the midpoint); the time error there is $O(\Delta\tau)$.

5 The linear system and matrix structure

Equation (2) written for the interior nodes $j = 1, \ldots, N-1$ becomes

$$A\,\mathbf{V}_{\text{int}}^{n+1} \;=\; B\,\mathbf{V}_{\text{int}}^{n} + \mathbf{b}_{\text{bnd}}^{n},$$

where A and B are tridiagonal $(N-1)\times(N-1)$ matrices with $(\beta_-, \beta_0, \beta_+)$ and $(\gamma_-, \gamma_0, \gamma_+)$ on their diagonals, and the boundary vector mixes both time levels:

$$\mathbf{b}_{\text{bnd}}^{n} = \begin{pmatrix} \gamma_-\,V_0^n - \beta_-\,V_0^{n+1} \\ 0 \\ \vdots \\ 0 \\ \gamma_+\,V_N^n - \beta_+\,V_N^{n+1} \end{pmatrix}. \tag{5}$$

Both lateral coefficients of the stencils contribute to the boundary correction, in contrast to BTCS where only the LHS coefficients appeared.

The per-step algorithm has three phases:

1. Apply the RHS stencil B to $\mathbf{V}_{\text{int}}^n$ as a three-point stencil sweep ($O(N)$, no need to assemble B explicitly).
2. Add the boundary correction $\mathbf{b}_{\text{bnd}}^n$ ($O(1)$: only two endpoints are non-zero).
3. Solve $A\,\mathbf{V}_{\text{int}}^{n+1} = (\text{result})$ by Thomas with the pre-factored A ($O(N)$).

Total cost per step: $O(N)$. Total cost: $O(N \cdot M)$, the same as BTCS.

Pre-factoring A. The matrix A is constant in time (again thanks to the constant-coefficient form of the transformed PDE). Factoring A once at setup and applying the factor at every step is the standard pattern, identical to BTCS.

6 Stability of the θ-scheme

The von Neumann amplification factor of (1), ignoring the reactive term, is

$$g(\xi) = \frac{1 - 4\alpha(1-\theta)\sin^2(\xi/2) + i\,(1-\theta)\beta_C \sin\xi}{1 + 4\alpha\theta\sin^2(\xi/2) - i\,\theta\beta_C \sin\xi}, \tag{6}$$

with $\beta_C := \mu\Delta\tau/\Delta x$, $\xi \in [-\pi, \pi]$ the Fourier wavenumber.

Proposition 6.1 (Stability of the θ-scheme)**.** *The θ-scheme is unconditionally von Neumann stable if and only if $\theta \geq 1/2$. For $\theta < 1/2$, $|g| \leq 1$ requires the conditional bound*

$$\alpha \;\leq\; \frac{1}{2(1-2\theta)}.$$

In particular, $\theta = 0$ gives $\alpha \leq 1/2$ (the FTCS CFL condition), and $\theta = 1/2$ is the smallest θ for which the bound is vacuous.

Sketch. Set $s = \sin^2(\xi/2)$, $u = \sin\xi$, so $u^2 = 4s(1-s)$. Then $|g|^2 = N(s,u)/D(s,u)$ where both numerator and denominator are quadratic in α. The condition $|g|^2 \leq 1$ becomes $D - N \geq 0$, which after expansion is

$$8\alpha s\left[(2\theta - 1) + 4\alpha\theta(1-\theta)s\right] \;+\; (\text{terms vanishing when } \beta_C = 0) \;\geq\; 0.$$

For $\theta \geq 1/2$, both bracketed terms are non-negative for any $\alpha \geq 0$, $s \in [0,1]$. For $\theta < 1/2$, the first term is negative and the bound becomes the stated inequality. (Full algebra in Strikwerda §6.3, or [4, §4.5].) □ □

CN at $\theta = 1/2$ is therefore *exactly on the boundary* between conditional and unconditional stability. This is the structural reason why CN, although stable, has weaker damping properties than BTCS, as we now examine.

7 The kink-induced oscillation problem

The amplification factor at the Nyquist wavenumber $\xi = \pi$ (highest representable frequency on the grid, $\sin^2(\pi/2) = 1$, $\sin\pi = 0$) is

$$g_\theta(\pi) = \frac{1 - 4\alpha(1-\theta)}{1 + 4\alpha\theta}.$$

For BTCS ($\theta = 1$) and CN ($\theta = 1/2$):

$$g_{\text{BTCS}}(\pi) = \frac{1}{1+4\alpha} \xrightarrow{\alpha\to\infty} 0, \qquad g_{\text{CN}}(\pi) = \frac{1-2\alpha}{1+2\alpha} \xrightarrow{\alpha\to\infty} -1.$$

BTCS damps the Nyquist mode exponentially; CN preserves it with a sign flip. In the regime of interest for CN ($\Delta\tau \sim \Delta x$, so $\alpha \sim 1/\Delta x \to \infty$ as $N \to \infty$), $|g_{\mathrm{CN}}(\pi)|$ is essentially 1, and the Nyquist mode is preserved indefinitely while alternating sign in time.

This is harmless for smooth initial data: the Fourier coefficient of the Nyquist mode in the initial vector V^0 is exponentially small, and "preserving exponentially small" yields nothing visible.

The payoff is not smooth. A vanilla call payoff $\phi(x) = K(e^x - 1)^+$ has a kink at $x = 0$: ϕ is continuous but ϕ' has a jump. The Fourier expansion of a function with such a kink has coefficients decaying only as $1/k^2$, not exponentially. The Nyquist coefficient in V^0 is therefore $O(1/N^2)$ — not exponentially small.

Consequence. An $O(1/N^2)$ amplitude at the Nyquist mode, preserved across M time steps with sign flip $g \approx -1$ at each, produces a visible oscillation pattern in the computed V^M:

- Bounded ($|g| \leq 1$, so the oscillation does not grow).
- Persistent (each step preserves the magnitude).
- Localised near the kink (where the kink-induced mode lives).
- Alternating sign at consecutive grid nodes (the spatial signature of $\xi = \pi$).

The visible artefact is a sawtooth of small amplitude near the strike, distinct from the explosive sawtooth of FTCS instability (see the FTCS note) by being bounded but visible. It degrades local accuracy near the strike: the price has $O(\Delta x)$ error there instead of $O(\Delta x^2)$, and the Greeks (gamma in particular) inherit the oscillation.

8 Rannacher time-stepping

The standard fix, due to Rannacher [5], is to replace the first *two* CN time steps by *four* half-time-step BTCS steps, each of length $\Delta\tau/2$:

$$\underbrace{\mathrm{BTCS}_{\Delta\tau/2} \circ \mathrm{BTCS}_{\Delta\tau/2} \circ \mathrm{BTCS}_{\Delta\tau/2} \circ \mathrm{BTCS}_{\Delta\tau/2}}_{\text{4 half-steps, total time } 2\Delta\tau} \circ \underbrace{\mathrm{CN}_{\Delta\tau} \circ \cdots \circ \mathrm{CN}_{\Delta\tau}}_{M-2 \text{ full steps}}. \tag{7}$$

Why it works. Each BTCS half-step damps the Nyquist mode by a factor $g_{\mathrm{BTCS},\Delta\tau/2}(\pi) = 1/(1+2\alpha)$ (α is the CN Fourier number; the half-step uses $\alpha/2$, giving $1/(1+2\alpha)$). After four half-steps the Nyquist mode is multiplied by $1/(1+2\alpha)^4$, which for $\alpha = O(N) \gg 1$ is $O(1/N^4)$ — well below the $O(\Delta x^2) = O(1/N^2)$ floor of the discretisation. By the time CN takes over, the kink-induced high-frequency modes are damped to round-off, and CN proceeds with effectively smooth data.

Why two full steps. Giles & Carter [6] prove that this is the minimum number that recovers $O(\Delta\tau^2)$ convergence in L^2 globally. For pointwise L^∞ convergence at the kink itself, four full steps of Rannacher would be needed; L^2 is sufficient for the price at any fixed spot, recovered by interpolation that smooths over the kink anyway.

Why BTCS, not finer CN. The artefact is not a function of $\Delta\tau$ being too large; it is the absence of damping. CN with arbitrarily small $\Delta\tau$ still has $|g_{\mathrm{CN}}(\pi)| = 1$ in the limit (the bound is sharp). Only a dissipative scheme can remove the irregular high-frequency content, and BTCS is the natural choice because the same Thomas factorisation can be reused with the appropriate coefficients.

Cost. Four half-steps of BTCS cost as much as two full CN steps. For $M \geq 10$ the overhead is below 5%; for $M \geq 100$ below 0.5%. Negligible compared to the convergence gain.

9 Implementation outline

The implementation consists of two modules:

- `quantlib.theta_scheme` (Python) / `quant::pde::theta_scheme_*` (C++): a generic θ-stepper parametrised by θ. Internally pre-factors the LHS matrix and runs a configurable number of time steps. Reusable as scaffolding for the American put (where the LHS solve is replaced by PSOR).
- `quantlib.cn` (Python) / `quant::pde::cn_*` (C++): the user-facing CN pricer for European options, with Rannacher smoothing enabled by default (`rannacher_steps=2`). Setting `rannacher_steps=0` gives vanilla CN, exposed to make the oscillation phenomenon testable.

  The signature mirrors BTCS:

```
cn_european_call(S, K, r, sigma, T, *, N, M, n_sigma=4.0,
                 rannacher_steps=2)
```

The θ-scheme module exposes a low-level kernel `theta_march(grid, V0, theta, bc_lower, bc_upper)` that the CN pricer calls twice for the Rannacher protocol: once with $\theta = 1$ for the initial half-stepped BTCS warm-up, then with $\theta = 1/2$ for the rest.

No replacement of FTCS/BTCS. The FTCS and BTCS modules do not use it. The θ-scheme infrastructure is a parallel mechanism intended for CN and any future generalisations.

10 Validation strategy

Six tests certify the new pricer:

1. Cross-validation against Black–Scholes closed form on a parameter grid (the standard battery used for FTCS and BTCS).
2. **Quadratic convergence in time:** halving $\Delta\tau$ at fixed (small) Δx should reduce the error by a factor of ≈ 4, not ≈ 2 as BTCS would. This is *the* test of CN.
3. **Balanced quadratic convergence:** doubling both N and M should reduce the error by ≈ 4, vs the ≈ 2 expected for BTCS in the same configuration.
4. **Vanilla CN (without Rannacher) has degraded convergence:** running the same problem with `rannacher_steps=0` should show a noticeably worse convergence ratio. This is the empirical evidence of the kink-induced oscillation.
5. Reproduction of BTCS via $\theta = 1$ in the θ-scheme: verifies the generic stepper against the BTCS pricer.
6. Put-call parity, input validation, etc.

The fourth test is the operational signature of the kink-induced oscillation, analogous to the CFL-violation diagnostic of FTCS: a configuration that produces visibly degraded output, captured as a passing assertion to prevent regression.

References

References

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